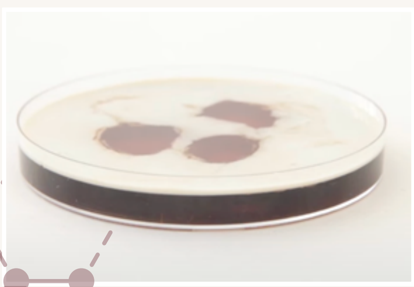
CONVECTION CURRENTS WITH COFFEE

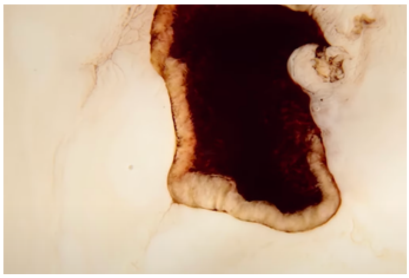
Lesson

The worksop discusses the concept of density and how substances of different densities interact with each other. For example, a denser substance has stronger forces between its particles than a less dense substance. If you were to pour the same volume of two liquids with different densities onto a counter, the less dense liquid would spread out more because the forces between its particles are weaker than its denser counterpart.

Experiment

The Convection Currents with Coffee experiment provides students with a dynamic demonstration of how substances of different densities can create currents with sideways particle movement. We will take a petri dish filled with coffee, ethanol, and sugar as well as a beaker filled with fat-enriched cream for the experiment. The sugar ensures that the net density of the coffee mixture remains high enough that if you carefully drop dots of cream onto the coffee's surface, the dots will float. When 80-90% of the coffee's surface is covered in cream, you can observe that the floating rafts of cream pull away in a sideways motion away from the holes of coffee. This is because of the ethanol in the coffee mixture, which has weaker forces between its particle than the cream does. When the cream on the surface moves sideways, other liquid (the coffee mixture) from under the surface is drawn up, creating tumbling rolls of cream that are caught in the convection. As the cream continues to pull away, the holes of coffee appear to move along the surface, creating beautiful patterns that illustrate the nature of convection currents.





IODINE CLOCK WITH VITAMIN C

Lesson

This worksop discusses the concept of reaction rates, catalysts, and concentrations. In particular, the lesson explains the components of a reaction and why a change in concentration will affect the reaction rate. For example, we use the following analogy for molecules of a substrate to help illustrate the reaction process: an empty room room is very easy to run around in, while a room full of people is hard to move around in and causes more collisions between people. Furthermore, the lesson touches on how temperature affects a reaction, supplemented by a demonstration performed by the workshop conductor.

Experiment

The Iodine Clock with Vitamin C experiment features two chemical reactions that are competing inside of a solution. We prepare two solutions (Sol A: Vitamin C, Iodine tincture, Distilled water; Sol B: Cornstarch, Hydrogen peroxide, Distilled water). Each lab group will be given different concentrations of Sol. B, which will all be poured by the students in their respective Sol A. at the same time. Combining the two colorless solutions will result in a sudden color change; however, each reaction will occur at a unique time. Students will then observe how differences in concentration affect reaction rates.